

Kartavya Soni

Florida, USA (Open for Relocation)

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EDUCATION

Master of Science in Computer Science

Florida Atlantic University, Boca Raton, FL

May 2026

GPA: 3.96

Bachelor of Engineering in Computer Engineering

Gujarat Technological University, Ahmedabad, India

May 2024

GPA: 3.70

RELEVANT EXPERIENCE

Data Science Intern

August 2023 – November 2023

Remote

- Architected **AWS S3/Glue pipelines** for semi-structured data, reducing missing values by **18%** via data validation.
- Optimized **10M+ records** using vectorized Pandas operations, reducing latency for downstream ML training.
- Engineered **7 features** and used **VIF analysis** to resolve multicollinearity, ensuring model stability and interpretability.
- Refined **Scikit-Learn classifiers** via **Stratified K-Fold CV** to handle class imbalance, achieving **0.94 F1-Score**.

Machine Learning Engineer

December 2023 – June 2024

Unified Mentor, Gurugram, India

- Developed a **Data Observability** framework for root-cause analysis, reducing production model incidents by **40%**.
- Architected **AWS SageMaker pipelines** to scale infrastructure and streamline continuous model deployment.
- Engineered robust **XGBoost** and **Scikit-Learn** classifiers, applying cross-validation for optimal model performance.
- Designed automated **MLOps workflows** with versioning and A/B evaluation gates to ensure reliability.

Governor, Student Government

May 2025 – April 2026

Florida Atlantic University, Davie, FL

- Led a **7-person team** using **Agile methodologies** to execute engagement strategies for **5,000+ students**.
- Optimized a **\$300K budget** via **time-series forecasting**, reducing fiscal variance by **8%**.
- Analyzed participation metrics** across 25+ events to isolate growth drivers, driving **20% YoY growth**.
- Streamlined administrative workflows** using digital tracking tools, reducing manual reporting time by **25%**.

PROJECTS

Global Lens: A Clearer View of Global Progress [Code](#)

Scikit-Learn, FastAPI, Streamlit

- Architected a **FastAPI** ETL pipeline to ingest **16k+** indicators, using a caching layer to reduce latency by **60%**.
- Built a country-segmentation engine using **PCA** and **K-Means** to identify 5 distinct global economic archetypes.
- Deployed **ARIMA** models with rolling-window cross-validation, improving GDP forecast accuracy (MAPE) by **12%**.
- Containerized a **Streamlit** dashboard via **Docker**, enabling real-time "What-If" scenario analysis on global clusters.

Loan Meter: Financial Risk Engine [Code](#)

XGBoost, Python, Pandas

- Engineered an **ETL pipeline** in **Pandas** for **200k+** records, automating outlier detection and missing data handling.
- Optimized an **XGBoost classifier** via **Bayesian Search**, achieving **0.76 ROC-AUC** and prioritizing high precision.
- Mitigated **10:1 class imbalance** using Cost-Sensitive Learning, improving minority class recall by **14%**.
- Leveraged **SHAP** to provide **Reason Codes** for loan denials, ensuring transparency and regulatory alignment.

CERTIFICATES

Google Advanced Data Analytics Professional Certificate – Google

May 2026

AWS Certified Cloud Practitioner – AWS

February 2026

SKILLS

- Languages & Core:** Python (Pandas, NumPy, Scikit-Learn), Advanced SQL, PySpark.
- Machine Learning:** Supervised Learning (XGBoost, Random Forest), Time-Series (ARIMA), Clustering (K-Means, PCA), Statistical Analysis (A/B Testing, Hypothesis Testing).
- MLOps & Cloud:** AWS (S3, Glue, SageMaker), Docker, Git/GitHub Actions, CI/CD Pipelines, MLflow, Pydantic.
- Systems & Visualization:** FastAPI (REST APIs), PostgreSQL, MongoDB, Tableau, Streamlit, Matplotlib, Seaborn.